QP. Code: 43203

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Third Semester BCA (SEP) Degree Examination
Dec./Jan. 2025-26
(Regular)
OPERATING SYSTEM

Time : 3 Hrs — Max. Marks : 80

Instruction to Candidates: Draw diagrams and write examples wherever necessary.

SECTION - A

Answer any TEN questions from the following. Each carries 2 marks. (10x2=20)

1. What is microkernel operating system?
2. Mention any four operating system services.
3. Define thread.
4. Mention the types of the scheduler.
5. What is race condition?
6. Mention any two classic problem of synchronization.
7. What is a page fault?
8. Define thrashing.
9. Define C-Look disk scheduling.
10. State any two differences between Directory and File.
11. Define context switch.
12. List the four necessary conditions for deadlock to occur.

SECTION - B

Answer any FOUR of the following, each carries FIVE marks. (4x5=20)

13. Explain different types of system calls with example.
14. Discuss multi threading models with diagram.
15. Consider the following set of processes with CPU burst time and arrival time given in milliseconds.

Process	Burst Time	Arrival Time
P1	6	0
P2	5	0
P3	8	0

Draw the Gantt chart and calculate the average waiting time and average turn around time using Round Robin (RR) scheduling. (Time quantum = 3)

16. Explain FCFS Disk scheduling algorithm with proper example and diagram.

17. Given memory partitions of 100 KB, 500 KB, 200 KB, 300 KB and 600 KB, allocate processes of size 212 KB, 417 KB, 112 KB and 426 KB using First Fit, Best Fit and Worst FIT. Show allocation results and calculate wasted memory.

SECTION - C

Answer any FOUR from the questions, each carries TEN marks. **(4x10=40)**

18. a) Explain the different types of operating system
    b) Explain in detail various structures of operating system.

19. a) What is PCB? Explain with a neat diagram.
    b) What is IPC? Explain the two IPC models in detail..

20. Consider the following system snapshot using data structures in the Banker's Algorithm, with resources A, B, C and D and processes P0 to P4.

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	4	0	0	1	6	0	1	2	3	2	1	1
P1	1	1	0	0	1	7	5	0				
P2	1	2	5	4	2	3	5	6				
P3	0	6	3	3	1	6	5	3				
P4	0	2	1	2	1	6	5	6				

Using Banker's algorithm, answer the following questions.

i) How many resources of types A, B, C and D are there?

ii) What are the contents of need matrix?

iii) If the system is in safe state, find the safe state sequence.

iv) If a request from process P4 arrives for additional resources of (A, B, C, D) = (1, 2, 0, 0), can the request be granted immediately?

21. a) Explain in detail segmentation with respect to memory management.
    b) Consider the following page reference string 1, 3, 0, 3, 5, 6, 3, 1, 6, 3 with 3 frames, find page faults using FIFO algorithm.

22. Discuss various File Allocation methods.

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